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Generative AI and the assurance of academic integrity in the context of South American higher education

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Purpose: This study focused on addressing generative Artificial Intelligence (AI) as a technologically and economically accessible practice for the majority of individuals, from the perspective of a recent innovation whose use has become widespread, including within the academic sphere of higher education. Accordingly, the study sought to establish the boundaries between educational support and academic dishonesty, which can only be defined through regulatory frameworks that determine such limits. From this standpoint, it is anticipated that, upon reviewing the legislation of South American countries, it was found that there is no specific regulation concerning the assurance of academic integrity.

Design/methodology/approach: The methodology employed in the research followed this sequence: the study was grounded in the post-positivist paradigm, addressing the existing regulations governing the use of AI; it adopted a qualitative approach due to the argumentative nature of the work, with an explanatory scope that makes it possible to establish the boundaries between academic support and academic dishonesty; the research technique used was documentary research, and the instrument applied was the documentary review guide.

Results: The general objective was to analyze the relationship between the use of generative AI and emerging forms of academic dishonesty in higher education, which call into question the assurance of academic integrity. To achieve this, it was necessary to address the adequacy of institutional responses, including: national regulations, governmental regulations, institutional regulations of universities, assessment practices, and integrity mechanisms such as anti-plagiarism systems or AI detectors.

Originality/value: Generative AI is a technological phenomenon of recent origin, for which there is little or no regulation, thereby creating room for academic dishonesty through the irresponsible use of AI by students within their educational process. In South America, there is no specific legislation governing the use of AI in higher education; however, there are notions that may be applied analogously. The results of the research show that it is necessary to establish clear parameters on the subject.

KEYWORDS

academic fraud, academic impersonation, academic integrity, authorship, regulatory framework

1 Introduction

The progressive incorporation of AI into educational systems constitutes one of the most influential technological phenomena of recent decades. In particular, the recent development of generative AI –capable of autonomously producing texts, arguments, images and other complex content– has begun to substantially transform teaching, learning and assessment processes in higher education. These technologies, initially conceived as tools for academic support, have acquired an operational capacity that directly affects knowledge production and the accreditation of university learning (Tan et al., 2025, p. 16).

In this context, higher education faces a structural challenge: ensuring academic integrity in an environment where the automated generation of content can replace students' cognitive effort and distort traditional assessment mechanisms. Academic integrity, understood as a guiding principle of the university system that guarantees the validity of knowledge, the quality of learning and the legitimacy of professional degrees, is grounded in fundamental pillars such as authentic authorship and intellectual honesty.

The unregulated use of generative AI places direct strain on these pillars, by facilitating practices that may result in apparent authorship, substitution of reasoning, and the improper appropriation of ideas without due attribution (Symeou et al., 2025, p. 299).

Despite the growing academic interest in the use of AI in education, a review of the international regulatory framework reveals a significant absence of specific regulations aimed at protecting academic integrity in the face of generative AI (Yusuf et al., 2024, p. 3). Relevant international instruments have developed general ethical principles regarding the responsible use of AI, digital literacy and the protection of fundamental rights; however, these texts omit a concrete treatment of academic dishonesty practices arising from the misuse of such technologies in the university context (Cacho, 2024, p. 17). This regulatory omission is particularly significant in higher education, where the assessment of learning and scientific production requires verifiable criteria of authorship and intellectual honesty.

The situation is even more problematic in the South American context. Regional regulation on AI applied to higher education is incipient, fragmented or non-existent, and in most countries is limited to public policy statements lacking binding force. With few isolated exceptions, there are no legal frameworks that expressly regulate the use of generative AI in the university sphere or establish normative mechanisms aimed at safeguarding academic integrity. This deficiency creates a scenario of legal uncertainty that affects both higher education institutions and lecturers and students, hindering the adoption of ethical and homogeneous practices in the educational process (Song, 2024, p. 22).

Against this backdrop, the present article aims to analyze the impact of the use of generative AI on academic integrity within South American higher education, with particular emphasis on the pillars of authentic authorship and intellectual honesty (Djoković et al., 2022, p. 4). Based on an examination of the applicable international and regional regulations, as well as

doctrinal analysis on academic integrity, the regulatory gap is identified and its implications for the university educational process are discussed, highlighting the need to develop normative frameworks that allow for the responsible and ethical use of these technologies (Surahman and Wang, 2022., p. 1,536).

The article is structured into several sections. First, it examines the development and use of generative AI in the academic sphere. Secondly, it analyses the international regulations applicable to AI in education, followed by a study of the existing South American regulatory framework. Subsequently, the concept of academic integrity and its fundamental pillars are addressed in the face of the emergence of generative AI. Finally, the results, discussion and conclusions are presented, in which the need to reform higher education regulations in South America is proposed, orienting the treatment of the unethical use of generative AI as a form of academic dishonesty comparable to plagiarism.

2 Methodology

This meta-analysis was conducted in accordance with recognized quality standards for the preparation and reporting of meta-analytical studies. The study is grounded in the post-positivist paradigm and adopts a critical realist perspective in order to examine the use of AI within the academic sphere, seeking to understand both the interpretations associated with this practice and the manner in which it is regulated by law. The research follows a qualitative approach, given its argumentative nature, and has an explanatory scope, as it seeks to establish the responsible use of AI that safeguards academic integrity, distinguishing it from academic dishonesty in its various forms. The research method is bibliographic and documentary, based on a systematic review of the legal sources regulating this subject. The principal research instrument was a documentary review guide, managed through the software Mendeley.

Because the study is based on documentary review, the search process was exhaustive, particularly with respect to South American countries, with the aim of identifying the legal provisions that most effectively safeguard academic integrity. In addition, the most relevant literature of recognized academic standing was collected in order to support the arguments developed in this study. An initial corpus of eighty references was examined from academic repositories such as Education Resources Information Center, PsycINFO, PubMed, Scopus, Publindex, EBSCO and SciELO, from which forty-three specialized sources were ultimately selected.

Initially, a systematic registration of sources obtained through the documentary research technique was carried out. Subsequently, the classification and detailed analysis of the essential concepts related to AI and academic integrity were undertaken. The inclusion criteria applied were as follows: 1. The time of publication, considering that the topic addressed is in constant expansion. 2. The academic relevance and authority of the authors. The analysis of all available international and South American regulatory frameworks in order to minimize potential bias. The exclusion criteria included: 1. Studies lacking objectivity under the current legal framework. 2. Studies based on self-reported data.

Using these criteria, a systematic bibliographic search was conducted across the selected databases without language restrictions, while excluding publications older than five years. Two reviewers identified and removed duplicate articles, which amounted to approximately eleven records. Subsequently, both reviewers examined a random subset in order to measure inter-reviewer agreement. Given the high level of concordance observed, one reviewer proceeded to assess the remaining articles for eligibility by analyzing their titles and abstracts. This screening process resulted in the exclusion of sixteen articles, as they consisted of review papers or commentaries.

Thereafter, the reviewers examined the full texts to ensure their eligibility. In total, forty-three articles were validated as supporting sources for this study. Finally, a cross-referencing process was employed to identify additional relevant studies, confirming that no significant publications had been excluded from the search. These theoretical elements served as the basis for drafting and structuring the results of the study. It is important to note that the research process revealed certain methodological limitations, particularly regarding the sources of positive law, which remain insufficiently developed in South America.

The selected documents were analyzed through an analytical, systematic, and interpretative reading of their content, with the aim of determining their relevance to the object of study. To this end, the documentary review guide made it possible to extract and organize the information according to previously defined categories, such as document type, normative or doctrinal nature, country of issuance, source hierarchy, treatment of AI, reference to academic integrity, and regulatory or argumentative scope. Subsequently, the information was coded and compared across the different sources in order to identify similarities, differences, regulatory gaps, and relevant doctrinal trends, thereby providing the basis for the construction of the study's results and discussion.

3 The use of generative AI in the academic context

AI is the product of an evolutionary process that begins with the ideas put forward by Alan Turing in 1950 through the Turing Test (Valor, 2024, p. 4). This process continued until the official emergence of the term AI in 1956 at the Dartmouth Conference (Sanabria et al., 2023, pp. 97–107). Within this period, significant milestones were recorded, such as Deep Blue defeating Kasparov in 1997.

Generative AI is a branch that focuses on the creation of new and original content based on models trained on large volumes of data (Ametic, 2024, p. 3). It is characterized by rapid adaptability, becoming a tool that promises greater efficiency and speed in the generation of documents, digital materials and even programming language code. A representative, though not exhaustive, history of generative AI may include some of the following milestones. In 1964, MIT computer scientist Joseph Weizenbaum developed ELIZA, a text-based natural language processing application. ELIZA was the first chatbot then referred to as a chatterbot, which used pattern-matching scripts to respond to natural language inputs.

In 1999, Nvidia launched GeForce, the first graphics processing unit. Originally developed to provide smooth motion graphics for video games, GPUs subsequently became the de facto platform for developing AI models and mining cryptocurrencies. In 2004, Google introduced its autocomplete function, which generates possible subsequent words or phrases as users enter their search terms. In 2013, the first variational autoencoders –VAE– emerged.

In 2014, the first generative adversarial networks –GAN– and diffusion models appeared. In 2017, Ashish Vaswani, a team from Google Brain and a group from the University of Toronto published “*Attention is All You Need*”, a paper that sets out the principles of transformer models, widely recognized as enabling the creation of foundational models and the most powerful generative AI tools currently under development. Between 2019 and 2020, OpenAI launched its large language models GPT Generative Pre-trained Transformer, GPT-2 and GPT-3.

In 2022, OpenAI introduced ChatGPT, a “*front-end*” of GPT-3 that generates complex, coherent and context form text in response to instructions from the end user. Following the notoriety and popularity of ChatGPT, which opened the floodgates, the development and release of generative AI products has proceeded at a rapid pace, including launches such as Google Bard –now Gemini–, Microsoft Copilot, IBM watsonx.ai, and Meta’s open-source large language model Llama-2.

Generative AI originated from a foundational “*deep learning*” model that serves as the basis for different types of applications. The most common at present are large language models –LLM–, which are a category of foundational models trained on vast amounts of data, enabling them to understand and generate natural language and other types of content to perform a wide range of tasks. There are also foundational models for image, video and music generation, as well as multimodal foundational models. The outcome of training a foundational model is a neural network of parameters, which is essentially a programmed or machine learning model that makes decisions in a manner similar to the human brain and is capable of autonomously generating content in response to inputs or instructions.

As scientific production on AI in education continues to grow, it is essential to conduct a comprehensive analysis of this literature in order to understand its implications and applications in the academic field (Tapullima et al., 2024, p. 3). “*The possibilities that AI brings to the analysis of scientific literature are wide considering all the repetitive tasks that the SLR methodology entails.*” (De la Torre et al., 2008, p. 2172).

By way of example of how the various types of generative AI may be used, a brief reference is made to their application in different university degree programmes, with the aim of highlighting the functions they perform, which range from argumentative support to academic reflection. It is prudent to indicate that the appropriate use of generative AI is necessary to complement the educational and research process; therefore, this study maintains that technology should be used as an academic support tool.

...the main ethical imperative is that students must complete university assignments based on the knowledge and skills they acquired during their university education unless their

instructors determine otherwise. Moral and legal responsibility in this regard always rests with human individuals. AI agents possess neither the legal standing nor the physical basis for moral agency, which makes them incapable of assuming such responsibilities (Dabis and Csáki, 2024, p. 10).

On the other hand, the principal risks to academic integrity will be proposed, ranging from the substitution of reasoning to the apparent authorship of academic work; therefore, it is undeniable that the unethical use of generative AI may seriously affect the educational process in higher education. It should be noted that this study does not oppose the responsible use of generative AI as academic support, but rather the absence of regulations that guarantee academic integrity.

Integrating emerging technologies, particularly Artificial Intelligence (AI), into university teaching and learning processes is becoming a crucial component of modern education. Generative AI models like ChatGPT have demonstrated significant potential in enhancing learning experiences by providing personalized, interactive, and adaptive education. However, the adoption of these advancements varies across regions due to differences in infrastructure, regulations, and resource access. This study

addresses the need for a tailored policy framework to facilitate the adoption of generative AI in higher education institutions (Alli et al., 2025, p. 1).

It should be emphasized that the table has solely demonstrative effects –it is not exhaustive–; what is truly important is that it shows that the use of generative AI may pose a risk to the educational process in higher education, calling academic integrity into question, since it could practically replace learning and the assessments that certify knowledge (Table 1).

Indeed, there is the larger issue of depending on GenAI for critical analysis, idea generation, and brainstorming. Some institutions appear to acknowledge the risks, though miss the mark, by reminding instructors to use activities that reinforce retrieval–implying that this is a skill that will undoubtedly atrophy as we strengthen our GenAI muscles (McDonald et al., 2025., pp. 9–10)

From this brief overview of the types of generative AI applicable to various university degree programmers, it is evident that the development of these technologies is extensive and continues to grow. Consequently, it may be argued that any academic work that should be produced by the student –with

TABLE 1 Types of generative AI used in higher education courses.

Area/Degree Program	Type of Generative AI	Generative AI tools that perform this function	Legitimate academic function	Main academic risk
Social Sciences and Humanities, Sociology, Education, History, Philosophy	Text generation	ChatGPT, Claude, Gemini, LLaMA-based chatbots	Argumentative support, outlines, theoretical analysis, comparison of approaches	Substitution of independent reasoning
Legal Studies, Law	Legal text generation	ChatGPT, Harvey AI, Lexis + AI, Westlaw AI	Legal methodology, case simulation	Apparent authorship, AI-assisted plagiarism
Economic and Business Sciences	Text+simulation	ChatGPT, Gemini, Copilot	Explanatory models, economic scenarios, conceptual analysis	Fabrication of data or conclusions
Engineering, Computer Science and IT	Code generation	GitHub Copilot, ChatGPT, Code Interpreter, Amazon CodeWhisperer	Programming learning, code drafts, debugging	Uncritical dependency, lack of understanding
Data Science/Statistics	Code+analytical output	ChatGPT, Python, Copilot, generative AutoML tools	Data exploration, model explanation, simulation	Non-replicable or unverifiable results
Architecture and Graphic Design	Image generation	Midjourney, DALL-E, Stable Diffusion	Visual prototyping, early-stage creative exploration	False artistic authorship
Digital Arts and Audiovisual Studies	Image/video/audio generation	Runway, DALL-E, Stable Diffusion, Sora	Experimental creation, aesthetic analysis	Presenting generated work as personal creation
Communication and Journalism	Text+image generation	ChatGPT, Gemini, DALL-E	Discourse analysis, campaign simulation	Informational manipulation
Marketing and Advertising	Persuasive text generation	ChatGPT, Jasper AI, Copy.ai	Strategy analysis, guided creativity	Authorship deception
Health Sciences	Textual simulation	ChatGPT, clinical simulation models	Hypothetical clinical cases, theoretical study	Improper diagnostic or clinical use
Education, Teacher Training	Pedagogical text generation	ChatGPT, Gemini	Instructional design, didactic reflection	Automation of pedagogical judgment

This table shows various types of generative AI and their specific use in university courses. (Table 1) Source: Author.

TABLE 2 International regulatory framework on the use of AI.

Instrument/Norm	Issuing body	Legal nature	Scope in education	Methodological contribution
Recommendation on the Ethics of AI	UNESCO	Soft law, international recommendation	Direct: education, research and training	Provides normative categories suitable for qualitative legal analysis and for building ethical assessment frameworks for the use of AI in education.
AI and Education: Guidance for Policy-Makers	UNESCO	Technical-policy document	Direct: educational systems	Contributes operational criteria for policy design; responses and to construct governance indicators for AI in education.
AI Act, AI Act	European Union	Hard law, binding regulation	Indirect: AI systems used in education	Offers a risk-based regulatory model; enables construction of analytical matrices on classification of systems, duties analogy to academic settings.
Ethical Guidelines on the Use of AI and Data in Teaching and Learning	European Commission	Soft law, ethical guidelines	Direct: teaching and learning processes	Supplies empirical variables: privacy, authorship, responsible use, assessment practices, in studies on academic integrity.
Framework Convention on AI	Council of Europe	International treaty, in adoption process	Indirect	Provides a human-rights-based analytical framework, and students' rights in educational uses of AI.
Global Digital Compact	United Nations	Global political declaration	Indirect	Useful for macro-contextual analysis of digital governance; comparative approaches to AI education.

This table shows regulatory developments in the international framework on the use of AI in academia. (Table 2) Source: Author.

the support of generative AI– could be supplanted by unethical practices that place their educational formation at risk (Table 2).

4 International regulatory framework on the use of AI in education

The most important text concerning the use of AI in education is the “*Recommendation on the Ethics of AI*”, developed by UNESCO in 2021, which devotes an entire “*Policy Action Area*” to this matter, entitled: “*Policy Action Area 8: Education Research*”. Within its provisions, it sets out the obligation of Member States to cooperate with “...international organisations, educational institutions, and private and non-governmental entities to provide people in all countries, at all levels, with appropriate knowledge about AI.” (UNESCO, 2021a, paragraph 101).

This highlights the importance of AI in the educational sphere, where technological means must always be regarded as academic support that facilitates knowledge acquisition. In the same vein, paragraph 103 establishes that: “*These programmes should be accessible to both technical and non-technical groups.*” (UNESCO, 2021a, paragraph 103), thereby conferring a universal character upon AI.

Reinforcing the importance of the use of AI in education, it is prudent to emphasize the notion of “*responsible and ethical use*”, which is reiterated throughout the text, distinguishing the use of AI from malpractice: “*AI should support the learning process without reducing cognitive capacities...*”, stressing that “...AI should not be subject to misuse, misappropriation...” (UNESCO, 2021a, paragraph 104).

In the same year, UNESCO issued the document entitled “*AI and Education: Guidance for Policy-Makers*”, which, although not a legal norm as such, explains concepts relating to the use of AI in education. It devotes sections to “*3.1 How can AI be harnessed to*

enhance education?” and “*3.2 How can AI be better harnessed for the common good in education?*” By way of summary, the document explains that AI should be used to promote collaborative learning, both for teachers seeking to improve teaching and for students who use its multiple tools as a means of learning.

Furthermore, section “*3.3 How can we ensure ethical, inclusive and equitable use of AI in education?*” is particularly relevant to this study, as it addresses issues such as the multiple risks of AI, algorithmic bias, the collection of personal data, and the intrusive nature of AI. However, the document does not address the issue of academic integrity or how it is affected by the use of AI (Table 3).

Continuing the review, brief reference is made to the “*European Union AI Act –AI Act–*”, formally Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonized rules on AI (European Parliament and of the Council, 2024). Although the text does not contain specific provisions governing the use of AI in education, it does include generally applicable rules and principles that emphasize the importance of AI literacy within education.

The document “*Ethical guidelines on the use of Artificial Intelligence –AI– and data in teaching and learning for educators*”, issued by the European Commission in 2022, constitutes a practical ethical guide aimed at teachers and educational decision-makers regarding the use of AI (European Commission, 2022). Its sections establish the ethical and appropriate use of AI, as well as the need to guide AI use towards various educational purposes.

Although the guide does not directly address the subject matter of this research, it is important to note that the responsible use of AI in the educational sphere would prevent malpractice and the resulting harm, which is unfortunately being evidenced in several studies. “*The harms of intensive technology use are varied and affect different physical, socio-*

TABLE 3 South American regulations on AI applied to education.

Country	Regulatory instrument national level	Includes education?	Legal nature	Methodological contribution to the research	Comparative value
Brazil	<i>Brazilian AI Strategy EBLA</i> , established by MCTI Ordinance No. 4,617/2021, and subsequent updates. Ministry of Science, Technology and Innovations.	Yes	National policy/ strategy	Provides an axis-based architecture, governance, training of human capital, RandDandI, ethics and regulation, allowing the operationalization of variables such as teacher training in AI, development of digital competencies, institutional adoption, and ethical criteria in education.	It allows comparison with a structured state policy model, useful for categories such as governance.
Colombia	<i>CONPES Document 4144</i> . National AI Policy. National Planning Department.	Indirect	Public policy, <i>soft law</i>	Useful for coding strategic pillars, goals, and mechanisms of inter-institutional governance; serves as a basis for constructing state capacity indicators.	It serves to contrast a model of state governance with programmatic goals.
Ecuador	<i>Pedagogical Guidelines for the Use of AI Tools in the Teaching–Learning Process</i> . Ministry of Education.	Yes	Technical guideline	Functions as a non-sectoral national reference for defining analytical categories related to pedagogical use, ethics, transparency, and non-substitution of teaching work, applicable by analogy to the analysis of higher education.	It constitutes the South American instrument most directly focused on educational practice, although with a lower normative rank than a law or national policy.
Perú	Law No. 31814. <i>Law promoting the use of AI in favor of the country’s economic and social development</i> . Congress of the Republic.	Yes	Law	Provides a solid dogmatic foundation, object, principles, and competent national authority, allowing the derivation of programmatic obligations and the analysis of their implementation in higher education through university regulations and policies.	It provides the clearest example of a national law in the region, useful for comparing normative density against mere guidelines or policies.
Argentina, Bolivia, Chile, Guyana, Paraguay, Suriname, Uruguay, Venezuela	Does not exist.	—	—	There are no regulations on this issue in the countries listed. It has been confirmed that there are bills under discussion.	It is advisable to classify it as the absence of an identified specific national instrument.

This comparison distinguishes between: laws; national policies or strategies; sectoral educational guidelines or programmers and absence of an identified specific national instrument. The review was updated to March 2026 on the basis of official sources available. (Table 3) Source: Author.

emotional and physiological dimensions” (Vázquez and Pascual, 2022, p. 106).

One of the most significant normative texts is the “*Framework Convention on AI and Human Rights, Democracy and the Rule of Law*”, an international instrument developed by the Council of Europe, issued in 2024 and open for signature by European States (Council of Europe, 2024). Its text establishes fundamental principles on the use of AI, and although it does not contain articles specifically related to education, it addresses in general terms the impacts of AI systems on human rights, among which education is included. “...the development of ethics-based AI considers human rights as an integral part of the discourse, albeit with varying degrees of importance” (Coddou et al., 2024, p. 357).

Finally, in 2024 the United Nations issued the declaration entitled “*Global Digital Compact*”, which does not have binding force; nevertheless, it constitutes an important text regarding the use of AI, particularly in education, as part of “*Objective 5. Enhance international governance of AI for the benefit of humanity.*” For the purposes of this study, paragraph 60 is cited:

We encourage the establishment of international partnerships to build AI capacity in order to develop education and training programmes, increase access to resources, including open AI models and systems, open training data and open

computing, and facilitate the training and development of AI models... (United Nations, 2024, paragraph 60)

Although international regulations governing the use of AI in education have been cited, in reality there is no substantial treatment of its use aimed at ensuring academic integrity, despite the efforts of the UN, UNESCO and the European Union to address a markedly novel issue. In this sense, it is anticipated at this stage that there is no international framework regulating the use of generative AI oriented towards the assurance of academic integrity in the field of education. “Worldwide, there has been virtually no research conducted, no guidelines agreed, no policies formulated, and no regulations enacted to address the specific ethical issues raised by the use of AI in education” (Holmes et al., 2018, p. 552).

As can be seen from the international regulatory framework on the use of AI in education, as well as from the bibliographic review of research studies, there is no regulation that limits or sanctions academic dishonesty arising from malpractice involving AI, which may take very diverse forms depending on the specialization and the type of AI used. Certainly, innovation and its regulation constitute a complex task.

A number of theories exist that seek to explain policymaking in organizations – for example, institutional theory (Meyer and Rowan, 1977) suggests that policy is influenced by external factors, such as laws, regulations and social norms. The external factor in the case examined here is a technological innovation of a kind that was not experienced by organizations, but is certainly of primary significance today (Wilson, 2025, p. 724)

5 South American regulatory framework on the use of AI in higher education

Perú: in South America, Peru is the only country that has specific legislation on the use of AI. In 2023, it enacted “Law No. 31814: Law Promoting the Use of AI in Favour of the Country’s Economic and Social Development.” Its text states that it is of national interest to promote the use of AI in multiple areas, expressly including, in Article 2: “...education and learning...” (Congress of the Republic of Peru, 2023).

This provision clearly encompasses higher education; however, it does not develop any regulations aimed at protecting academic integrity. In fact, the statute consists of only five fairly general articles. Possibly the only provision related to the subject of this study is Article 4, which designates the Secretariat of Digital Government and Digital Transformation as the body responsible for directing, evaluating and supervising the use and promotion of AI development. Nevertheless, this mandate has not been implemented, as no additional regulations have been enacted.

Colombia: through its National Planning Department – DNP–, issued in 2025 the document entitled “CONPES No. 4144: National AI Policy” (Colombia National Planning

Department - DNP, 2025). This is not a law, but rather a public policy guideline that establishes strategies to guide the development, adoption, regulation and ethical implementation of AI. Although this public policy does not regulate specific aspects of education, its text refers to the State’s obligation to strengthen capacities in educational and research institutions, as well as to incorporate pedagogical and technical competencies in AI.

Brazil: through its *Ministério da Ciência, Tecnologia e Inovações*, developed in 2021 the “Brazilian AI Strategy (EBIA)”. This is not a legal norm, but a set of public policies guiding State action, primarily with regard to “boosting scientific research in AI”. It is worth noting that this guideline expressly recognizes higher education institutions as “...central spaces for the training of AI talent...” and as “...responsible for developing ethical and critical capacities regarding the use of AI.” (Brazil, Ministry of Science, Technology and Innovations, 2021).

Ecuador: presents a similar case through the public policy guidelines entitled “Pedagogical Guidelines for the Use of AI Tools in the Teaching–Learning Process”, prepared by the Ministry of Education of Ecuador in 2024. Although the document is mainly aimed at general education, its content may be extended by analogy to higher education, especially in the absence of specific State university regulations on AI.

The Ecuadorian text is particularly relevant for analyzing academic integrity, authorship and the limits that should govern the use of generative AI in higher education, as it establishes the following Guiding Principles: “Responsible and ethical use of AI”, “Non-substitution of the teaching role or the student’s cognitive effort”, “Critical thinking and human supervision”; and “Respect for originality and authorship” (Ministry of Education of Ecuador, 2024).

Although Chile does not yet have any regulatory framework governing the use of AI, it should be noted that there is a bill developed in 2024 by the Ministry of Science, Technology, Knowledge and Innovation entitled “Bill Regulating AI Systems (Bulletin No. 16821-19)” (Ministry of Science, Technology, Knowledge and Innovation of Chile, 2024). Unfortunately, it is still under discussion, as is the case in Argentina with the so-called “National AI Bill”, developed by the Congress (National Congress of Argentina, 2025).

On the other hand, the remaining South American countries do not have specific regulations on the use of AI in the educational sphere, much less its application in higher education or with regard to academic integrity. “The primary concerns regarding the use of GenAI in higher education have been the issues of academic integrity and academic dishonesty” (An et al., 2025, p. 3). This regulatory gap has been identified in countries such as Bolivia, Paraguay, Uruguay and Venezuela.

From the evidence gathered, it may be argued that in South America there is virtually no regulation governing the use of generative AI in the academic sphere. The limited regulations developed by countries such as Peru are incipient; therefore, in practice, no South American country has specific regulation in this regard. In this sense, the importance of urgently developing regulations that allow for the responsible and ethical use of these technologies in the educational sphere is underscored, with the aim of ensuring academic integrity. It is nevertheless fair to note that, upon reviewing international regulations, it was

also found that there are no clear regulatory frameworks worldwide.

With existing guidance at institutional level, faculty feel that current guidelines are difficult to enforce, especially regarding practicality and clarity. Ning stated, ‘I haven’t read any official guidance. I know there were discussions, but I don’t think we’ve received anything practical that actually tells us how to integrate AI appropriately’ (Jiang et al., 2025., p. 8)

As can be seen from the cited passage, in addition to the fact that the use of generative AI in the academic sphere has not been adequately regulated—thereby creating tension with academic integrity—the absence of regulation prevents lecturers and students from being able to use these technological tools correctly within the educational process. For this reason, regulating this issue also entails improving these practices and facilitating their ethical use.

For a better understanding of the subject, it should be noted that Argentina, Bolivia, Chile, Guyana, Paraguay, Suriname, Uruguay, and Venezuela do not have any kind of regulatory framework on the use of AI in higher education, let alone one related to ensuring academic integrity in the face of this phenomenon. Despite this, it has been found that some of these countries, such as Chile, have bills under debate.

6 Legal aspects of the use of AI in higher education

As can be determined from the “*International regulations on the use of AI in education*”, there are certain declarations lacking binding force that address the use of AI within the sphere of higher education. These instruments also express positions regarding the ethical implications of such use; however, there is no binding legal provision that requires verification of their compliance as part of governance mechanisms in higher education. “*The absence of a binding international legal framework on AI poses significant challenges, including regulatory gaps, ethical dilemmas, risks of misuse, and uneven governance across jurisdictions.*” (Saidi, 2025, p.254)

The normative frameworks developed by the United Nations Educational, Scientific and Cultural Organization may be applied by South American countries, given that it is a specialized agency of the United Nations system. Nevertheless, the remaining regulations cited do not apply to South American countries, as they have been issued by the European Union, the European Commission, or the Council of Europe.

Despite this, it is important to note that all these regulatory instruments merely establish general criteria concerning governance in education, without containing provisions aimed at verifying compliance with the ethical use of AI in higher education. “*This is why, governments are pushing towards AI regulation domestically especially given the need to balance innovation societal protection via laws.*” (Zaidan and Ibrahim, 2024, p.9)

With regard to the “*South American regulatory framework on the use of AI in higher education*”, it was observed that only a few South American countries have developed an incipient regulatory framework concerning the ethical use of AI in higher education. Conversely, it was found that in the vast majority of South American countries there is no normative development on this matter, although it should be noted that some states are currently drafting legislative proposals.

On this basis, it can be argued that, within the South American context, the ethical use of AI in higher education is not effectively verified through regulatory mechanisms. “*Furthermore, fostering interdisciplinary collaboration and engagement with diverse stakeholders, including academia, industry, government, civil society, and international organizations, is essential to drive research and policy in AI.*” (Igbinenikaro and Adewusi, 2024, p.501)

In order to appropriately regulate the ethical use of AI in higher education, it is essential for the law to establish mechanisms for its verification, thereby preventing such use from undermining academic integrity, which constitutes a fundamental guarantee of the educational process. In the absence of specific regulatory provisions, some universities have implemented institutional mechanisms to verify the ethical use of this technology. One example is the adoption of the system Compilatio, developed in France by the company EdTech, which is capable of detecting the use of AI with an accuracy rate of approximately 99%.

It is therefore reiterated that, although some universities may establish this type of provision within their internal regulations, they do not do so under the protection of a state-level legal framework that determines its mandatory nature, ensuring the proper functioning of higher education institutions in general and safeguarding academic integrity. The legal issue addressed in this study is to establish the need to develop a regulatory framework in this regard.

7 Academic Integrity

Although this study does not seek to delve deeply into pedagogical concepts, it considers it important to refer to certain notions that are strictly related to higher education. Academic integrity constitutes a commitment to honesty that safeguards the educational process and seeks to ensure the quality of learning; it thereby confers validity upon academic degrees and the professional practice of students. For this reason, academic integrity represents a guiding principle of the university system and gives legitimacy to academic knowledge.

The analysis of the impact of academic integrity on the quality of education has shown its direct connection with the development of critical thinking, independence, responsibility and lifelong learning among students. Failure to comply with the principles of integrity, on the contrary, leads to a decrease in the level of knowledge, devaluation of diplomas and undermining the authority of higher education institutions (Liuta et al., 2025., p. 104)

Within classical literature, academic integrity is grounded in four pillars: authentic authorship, intellectual honesty, the student's academic responsibility, and institutional guarantees in the assessment process. Although all of these pillars are indispensable for ensuring academic integrity, particular emphasis in this study is placed on “*authentic authorship*” and “*intellectual honesty*”, since the use of generative AI directly impacts upon them.

“*Authentic authorship*” is the essential element of academic integrity through which the student assumes, understands and is able to rationally defend the intellectual parentage of the work submitted, ensuring that the academic product reflects a formative process that can be verified, even when it has been supported by sources, tools or auxiliary technologies. Its antithesis is apparent authorship, which occurs when a student presents the work of another as their own.

An effective tool for ensuring the quality of higher education is academic integrity, which is recognized as the moral code and ethical rules of the educational, scientific, and civilized community. According to the results of the project Impact of Policies for Plagiarism in Higher Education Across Europe in 27 EU member states, the systems and methods of combating plagiarism were analyzed and recommendations were made regarding the need to popularize the institutional culture of academic integrity (Zabiiaka et al., 2023., p. 151)

To achieve authentic authorship, the intervention of three subjects in the creation of knowledge is indispensable: the student –author–, the academic author, and the pedagogical tutor. Authentic authorship depends on the involvement of these three roles in activities related to the educational process, validating it through supervised learning that ensures the student's work. Subordinating authentic authorship entirely to the student may lead to the transgression of this pillar.

The purpose of academic tutoring requires a process of reflection around the educational goal in which three actors intervene to facilitate the teaching and learning process. In order to achieve the expected academic training objectives, it is necessary that there be clarity regarding the functions established for each role, as this involves coordination between the academic author and the pedagogical tutor, as well as responsibilities on the part of the student. This requires engagement with motivation and ethics that enables the construction and reconstruction of knowledge, contributing to academic training, learning and the student's professional role (Núñez-Naranjo, 2021., p. 72)

In the same vein, the pillar of “*intellectual honesty*” is proposed, understood as the ethical-academic duty of the student not to falsify the reality of their cognitive process, whether in the use of sources, the presentation of results, or the attribution of intellectual merit. Intellectual honesty refers to not altering the results of academic work and to maintaining strict objectivity. It differs from authentic authorship in that the latter only ensures the real authorship of the work.

The virtue of academic honesty is crucial in higher learning and ensures that students engage in honest learning and uphold honesty in their academic endeavors. Different types of academic dishonesty, including test cheating, plagiarism, unauthorized help, faking information, and misrepresentation on schoolwork, continue to be a massive problem in schools all over the world (Shah et al., 2025., p. 243)

The emergence of generative AI does not eliminate these pillars, but it places them under strain, requiring a redefinition of how they can be guaranteed. Unfortunately, as has been established, there is no regulation worldwide governing the use of generative AI in education, much less in higher education, and even more concerning, there are no parameters that allow academic integrity to be ensured within the educational process. This therefore constitutes an urgent issue.

In light of this, it becomes clear that the pillars of higher education “*authentic authorship*” and “*intellectual honesty*”, are not being verified in the context of the use of generative AI. Consequently, there is no certainty that academic integrity is being guaranteed, due to the absence of regulation in South America governing this matter within higher education.

Finally, it is necessary to distinguish the irresponsible or unethical use of technology, conduct which, despite not being regulated, could be construed as a form of academic dishonesty. Academic dishonesty is understood as dishonest behavior by the student that is usually sanctioned by the university —without considering generative AI— and is generally understood as copying, plagiarizing or impersonating within the educational process, thereby transgressing academic integrity and Higher Education.

What constitutes dishonest behavior needs to be clearly defined generally by university or college honor codes and defined in specific terms by course instructors. Dishonest behavior in both the face-to-face and online settings results in the student gaining an unfair advantage which misrepresents their true knowledge and abilities. The new opportunities and challenges pertaining to academic dishonesty in the online environment are due in large part to the absence of physical proctoring of course work. This can lead to student identity issues, the use of unauthorized materials or assistance, and the perception that there is a lower risk of getting caught (Adzima, 2021., p. 489)

Emphasizing that such conduct is not currently regulated in South America, this study proposes that it should be sanctioned as plagiarism, since it arises from the appropriation of ideas generated by generative AI, which in turn derive from the ideas of real authors who publish their work on the internet, and which are not cited by the student.

Plagiarism is not merely an academic transgression; it is a profound violation of the core principles of truth, originality, and integrity that underpin the academic world. The implications of plagiarism are extensive, affecting not only the individuals directly involved but also the very

foundation of our educational system. Academic institutions, designed to nurture knowledge and character development, find their principles shaken by the insidious presence of plagiarism. It is vital to understand the multifaceted causes and motivations behind this pervasive issue to effectively combat it (Mulenga and Shilongo, 2024., p. 41)

8 Results and discussion

AI, from the Turing Test to contemporary generative AI capable of creating content, has undergone a long process of development. Its evolution, from ELIZA to transformer models and ChatGPT, has transformed education; while its proper use may enrich the educational experience, its inappropriate use threatens academic integrity. This brief overview has made it possible to identify the wide range of technologies applicable to different university degree programmers, which makes it necessary to ensure that their use does not replace students' learning, assessment and research activities.

The “*Recommendation on the Ethics of AI*” issued by UNESCO (2021a, b) is the most relevant instrument concerning the use of AI in education and research, as it highlights its value as a tool for academic support. The document addresses risks such as algorithmic bias, technological intrusion and data collection, but omits the impact on academic integrity, as AI may be used to substitute human reasoning and to generate academic work presented as one's own. A similar omission can be observed in the “*European Union Artificial Intelligence Act –AI Act–*” and in the “*Ethical guidelines on the use of Artificial Intelligence –AI– and data in teaching and learning for educators*” issued by the European Commission. The United Nations' “*Global Digital Compact*” also fails to regulate this aspect.

In South America, regulation on the use of AI in higher education is virtually non-existent. Peru has specific legislation, although it is limited to five articles of a general nature. Brazil, Colombia and Ecuador have guidelines that function as orientate public policies, without binding normative force. Argentina and Chile lack enacted legislation, although they have presented draft bills that have not yet been approved. The remaining South American countries do not have regulations, draft legislation or public policies focused on the application of AI in higher education, much less on the protection of academic integrity.

Without intending to theorize on pedagogy, it was nevertheless necessary to address concepts related to higher education, such as academic integrity, a principle that underpins education itself. Academic integrity rests on four pillars: authentic authorship, intellectual honesty, the student's academic responsibility and institutional guarantees in the assessment process. In the contemporary context, with the advent of generative AI, the principles of authentic authorship and intellectual honesty are being challenged, casting doubt on the veracity of knowledge and the legitimacy of higher education.

In summary, the principal finding of this research is that the state regulatory frameworks of South American countries do not guarantee the ethical use of AI, which should be oriented toward safeguarding academic integrity as a fundamental element of the educational process. The absence of specific state

regulation establishing control mechanisms to ensure academic integrity calls into question the quality of higher education in South American countries.

The irresponsible use of technology represents conduct that may be interpreted as academic dishonesty, even though it is not expressly regulated in international or South American legal frameworks. This practice entails the improper appropriation of ideas which, although generated by AI, originate from the work of real authors available on the internet and are not being cited. For this reason, it is proposed that this type of behavior be equated with plagiarism and sanctioned as such, in order to preserve the pillars of academic integrity.

9 Conclusions

There is a wide variety of generative AI models that are applicable to numerous university degree programmers; therefore, their use as a means of academic support must be regulated.

International regulations address the use of AI in extremely broad terms and, although they refer to its orientation towards education and research, they do not in fact develop parameters that determine how lecturers and students may appropriately incorporate these technologies while ensuring academic integrity.

Regulation in South America is virtually non-existent, as is the case with State-issued guidelines establishing public policies on the use of generative AI; it should also be noted that several countries are only at the stage of debating draft legislation on this matter.

It may be argued that the use of generative AI in higher education lacks regulatory frameworks and that, for this reason, its practice may be questioned, since academic integrity as a guiding principle of the educational process is not being ensured, thereby allowing room for acts of academic dishonesty.

Due to the absence of regulation, the pillars of academic integrity “*authentic authorship*” and “*intellectual honesty*”, cannot be verified. In this sense, it is argued that the educational process in higher education is not being guaranteed.

This study recommends reforming higher education regulations in South America, with the purpose of equating the unethical use of generative AI with plagiarism and sanctioning it in the same manner.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

JC: Writing – original draft, Writing – review & editing. RL: Writing – original draft, Writing – review & editing. LB: Writing – original draft, Writing – review & editing.

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The author(s) declared that generative AI was not used in the creation of this manuscript.

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